

Supplementary Materials: Building Resource-Efficient Structured Event Extraction Systems with Small Language Models

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A Supplementary Tables and Schemas

A.1 Event Schema

The extraction schema uses a JSON Schema *oneOf* discriminator to switch on *event_type*, which has six possible values.

Table 1: Shipment Delay Event Schema

Field Category	Associated Fields / Constraints
Required Fields	carrier, reason
Optional Fields	consignment_description, origin, destination, delay_duration_days
Constraints	None

Table 2: Supplier Insolvency Event Schema

Field Category	Associated Fields / Constraints
Required Fields	supplier_name, legal_action
Optional Fields	filing_date, jurisdiction, affected_product_categories
Constraints	legal_action: Bankruptcy, Liquidation, Restructuring, Receivership, Insolvency

Table 3: Force Majeure Event Schema

Field Category	Associated Fields / Constraints
Required Fields	declaring_entity, cause
Optional Fields	location_affected, effective_date
Constraints	None

Table 4: Facility Halt Event Schema

Field Category	Associated Fields / Constraints
Required Fields	operator, facility_location, disruption_type
Optional Fields	start_date, expected_restart_date
Constraints	disruption_type: Strike, Accident, Utility_Outage, Maintenance, Natural_Disaster, Cyberattack, Geopolitical_Conflict, Regulatory_Halt, Labor_Dispute

Table 5: Tariff Change Event Schema

Field Category	Associated Fields / Constraints
Required Fields	originating_country, target_country, tariff_action
Optional Fields	tariff_percentage_increase, affected_goods, effective_date
Constraints	tariff_action: Increase, Decrease, Implementation, Exemption, Elimination

Table 6: No Event Schema

Field Category	Associated Fields / Constraints
Required Fields	event_type (= "NoEvent")
Optional Fields	-
Constraints	-

A.2 Qualitative Examples (Data)

Example A: Extraction (Facility Halt – UAW strike)

Table 7: Gold Target Values (Example A – UAW Strike)

Field	Gold Target Value
event_type	Facility Halt
source_timestamp	2023-09-15T00:00:00Z
text_evidence	"The situation forced GM to implement an immediate facility halt at the assembly and stamping plant"
operator	"General Motors Company"
facility_location	"Wentzville Assembly plant, Missouri, USA"
disruption_type	Strike
start_date	2023-09-15T00:00:00Z
expected_restart_date	2023-10-30T00:00:00Z

Table 8: Pipeline Prediction (Example A – UAW Strike)

Field	Pipeline Extracted Value
event_type	Facility Halt
source_timestamp	2023-09-15T00:00:00Z
text_evidence	“The situation forced GM to implement an immediate facility halt at the assembly and stamping plant”
operator	“General Motors” (<i>Near-Match</i>)
facility_location	“Wentzville Assembly Plant, Wentzville, Missouri” (<i>Near-Match</i>)
disruption_type	Strike
start_date	2023-09-15T00:00:00Z
expected_restart_date	2023-10-30T00:00:00Z

Example B: Misclassification – Class Collapse (ShipmentDelay to FacilityHalt)

Table 9: Gold Target Values (Example B – Shipment Delay)

Field	Gold Target Value
event_type	Shipment Delay
text_evidence	“manual processing slow-down caused a significant shipment delay of exactly 6 days”
carrier	Ocean Network Express
delay_duration_days	6

Table 10: Pipeline Prediction (Example B – Class Collapse to Facility Halt)

Field	Pipeline Extracted Value
event_type	Facility Halt ×
text_evidence	“Ocean Network Express... had to wait at anchorage while port workers manually processed incoming containers”
carrier	- (<i>Wrong Schema</i>)
delay_duration_days	- (<i>Wrong Schema</i>)

Example C: Schema failure in baseline (structural inconsistency)

While it is possible to extract event types such as *Supplier Insolvency* or *Facility Halt* from the baseline model’s responses, it does not provide structured responses consistently. The test run indicated that the baseline only correctly predicted 7 out of 30 event types (23.33%) with sufficient schema validity. In 23 out of 30 cases, the baseline’s output contained structural validity issues, such as missing arguments or invalid JSON formatting.

A.3 LoRA Configuration

Table 11: Complete LoRA and Training Configuration

Parameter	Value	Source
Base model	Qwen/Qwen2.5-1.5B-Instruct	adapter_config.json
Architecture	Qwen2ForCausalLM, 28 layers, hidden_size=1536, 12 attn heads, 2 KV heads (GQA), intermediate_size=8960	config.json
Rank (r)	16	adapter_config.json
Alpha (α)	32	adapter_config.json
Scaling factor (α/r)	2.0	derived
Dropout	0.1	adapter_config.json
use_dora / use_rslora	false / false	adapter_config.json
PEFT version	0.19.1	model card [?]
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj	adapter_config.json
Base model params	1,543,714,304 ($\sim 1.54\text{B}$)	computed
Trainable params	18,464,768 ($\sim 18.46\text{M}$)	computed
LoRA % of base	1.196%	computed
Training dtype	bfloat16 (GPU) / float32 (CPU)	train_qwen_lora.py
Inference dtype	float16 (GPU) / float32 (CPU)	trriage_pipeline.py
Optimizer	AdamW [?], lr=1e-4	train_qwen_lora.py
Epochs	3	train_qwen_lora.py
Batch size	1 (Qwen), 8 (DistilBERT)	training scripts
Max sequence len	1,024 tokens (Qwen), 512 tokens (DistilBERT)	training scripts
DistilBERT dropout	0.2	train_distilbert.py
Random seed	42	both training scripts
Max new tokens	512	trriage_pipeline.py
Decoding strategy	Greedy (do_sample=False)	trriage_pipeline.py
Adapter file size	73.9 MB (adapter_model.safetensors)	filesystem

A.4 LoRA Parameter statistics

Table 12: LoRA Parameter statistics

Module	Projection Dim	LoRA Matrix A size	LoRA Matrix B size	Params per layer
q_proj	1536	16×1536	1536×16	49,152
k_proj	256	16×256	256×16	8,192
v_proj	256	16×256	256×16	8,192
o_proj	1536	16×1536	1536×16	49,152
gate_proj	8960	16×1536	8960×16	167,936
up_proj	8960	16×1536	8960×16	167,936
down_proj	1536	16×8960	1536×16	167,936

A.5 Domain Generalization

The proposed pipeline is domain agnostic and can be adapted to other enterprise information extraction tasks by replacing the event taxonomy, annotation guidelines, and output schema while preserving the overall architecture. The following examples illustrate this adaptation.

A.5.1 Financial Market Intelligence

The triage classifier is fine-tuned to identify market-moving news with high recall, while the SLM extracts structured financial events using the same schema-constrained decoding and validation pipeline. Representative event types include *Trading Halt*, *Regulatory Fine*, *Earnings Miss*, *Merger and Acquisition*, and *Executive Leadership Change*.

For example,

“Singapore’s Samlit Moneychanger Pte. Ltd., along with its director and compliance manager, is facing court charges for failures related to complaint handling and obstruction of justice during investigations.”

is transformed into a structured event similar to:

```
{
  "event_type": "RegulatoryFine",
  "arguments": {
    "company_name": "Samlit Moneychanger Pte. Ltd.",
    "regulatory_body": "Singapore Court / MAS",
    "reason": "Complaint handling failures"
  }
}
```

A.5.2 Healthcare Event Intelligence

The same architecture can be applied to pharmacovigilance and clinical event monitoring by retraining the triage classifier on medication-related documents and adapting the extraction schema. Representative event types include *Adverse Drug Reaction*, *Medication Error*, *Drug Interaction*, *Allergic Reaction*, and *Treatment Discontinuation*.

For example,

“Patient (ID: 9482) was administered Lisinopril 20 mg daily for hypertension. Within three days, the patient developed a persistent dry cough requiring discontinuation of the medication.”

is transformed into:

```

{
  "event_type": "AdverseDrugReaction",
  "arguments": {
    "patient_id": "9482",
    "received_medication": "Lisinopril",
    "adverse_reaction": "Dry cough"
  }
}

```

Both examples reuse the same engineering methodology presented in this paper. Only the domain-specific event taxonomy, annotation guidelines, and structured output schema are modified, while the staged inference pipeline, parameter-efficient fine-tuning, schema-constrained decoding, and hallucination validation remain unchanged.

B Supplementary Analysis and Discussion

B.1 Training Progression (Validation)

Table 13 summarizes the validation performance across the three fine-tuning epochs. While the training loss decreased consistently throughout training, the largest improvement in extraction quality occurred between Epochs 2 and 3. Top-Level F1 increased from 52.05% to 76.94%, whereas Argument-Level F1 improved substantially from 37.56% to 68.35%.

Table 13: Training Loss and Performance Progression

Epoch	Train Loss	Top-Level F1	Arguments F1
1	0.1923	34.21%	36.38%
2	0.0492	52.05%	37.56%
3	0.0114	~76.94%	~68.35%

This progression suggests that event classification converges earlier than argument extraction. Although the model learned to identify event types after two epochs, an additional epoch was required to reliably extract event arguments. Consequently, selecting the final checkpoint based solely on training loss or event classification performance would have resulted in inferior structured extraction quality.

B.2 Qualitative Examples (Discussion)

Example A (Success Case): Demonstrates robust structured extraction, including semantic normalization (e.g., “General Motors Company” → “General Motors”) and ISO 8601 date formatting.

Example B (Systematic Failure): Illustrates class collapse, where the model predicts the dominant *Facility Halt* class instead of *Shipment Delay*, resulting in missing schema-specific arguments.

Example C (Baseline Comparison): Demonstrates the importance of FSM-constrained decoding. Although the zero-shot model often predicts the correct event type, it produces invalid structured outputs in 76.6% of cases.

Complete inference examples are provided in Appendix A.2.

B.3 Per-class prediction analysis

The prediction distribution indicates a systematic bias toward the *Facility Halt* class at the expense of *Shipment Delay*, supporting the class ambiguity discussed in Appendix B.6.4. The

remaining event types exhibit balanced prediction frequencies. Table 14 below shows the difference between gold and predicted counts for each event class on the test set.

Table 14: Per-class Prediction Delta

Event Type	Gold	Predicted	Delta	Note
Facility Halt	6	12	+6	Over-predicted
Supplier Insolvency	6	6	0	—
Force Majeure	6	6	0	—
Tariff Change	6	6	0	—
Shipment Delay	6	0	-6	Under-predicted

B.4 LoRA Adapter Weight Analysis

B.4.1 Parameter statistics

The LoRA adapters introduce 18.46M trainable parameters distributed across all 28 transformer layers. Detailed parameter statistics are provided in Appendix A.4.

B.4.2 Domain adaptation evidence

The adapter update magnitudes in Table 15 indicate that fine-tuning primarily modifies the feed-forward (MLP) projections, while comparatively smaller updates are applied to the attention projections. This observation suggests that domain adaptation is achieved predominantly through transformations within the MLP blocks rather than by substantially altering the attention mechanism.

Table 15: Mean Adapter Update Magnitude (Frobenius Norm) by Module

Module	Mean Magnitude
gate_proj (MLP)	0.83
up_proj (MLP)	0.71
down_proj (MLP)	0.32
q_proj (Attention Query)	0.29
o_proj (Attention Output)	0.28
k_proj (Attention Key)	0.11
v_proj (Attention Value)	0.11

B.5 Domain Generalization: Broader Applicability

The proposed two-stage pipeline is not specific to supply chain event extraction. Its core architecture, including document triage, parameter-efficient fine-tuning, schema-constrained decoding, and selective hallucination validation can be reused across enterprise information extraction tasks with only domain-specific adaptations. The components requiring modification are:

- Document corpus and training dataset

- Event taxonomy
- Annotation guidelines
- JSON schema
- Prompt templates
- Domain-specific evaluation criteria

To illustrate this transferability, Appendix A.5 presents conceptual adaptations of the proposed methodology to financial market intelligence and healthcare event extraction.

B.6 Lessons Learned and Future Directions

B.6.1 Prompt-Annotation Alignment

Prompt instructions, annotation guidelines, and output schemas must remain consistent throughout the pipeline. Misalignment reduces schema consistency and extraction quality. Future work should investigate automated methods for maintaining this alignment.

B.6.2 Managing Error-Cost Asymmetry

False negatives during document triage are substantially more expensive than false positives because discarded documents never reach the extraction stage. Consequently, optimizing the triage classifier for high recall proved more effective than maximizing precision. Future work may explore deployment-specific calibration techniques such as Platt Scaling [?].

B.6.3 Hallucination Detection Requires Field Awareness

Lexical grounding is effective for free-text fields but insufficient for categorical attributes constrained by predefined schemas. Future work should investigate field-aware confidence estimation and semantic grounding beyond lexical matching.

B.6.4 Event Taxonomy and Class Ambiguity

Balanced datasets alone cannot resolve semantically overlapping event classes. The confusion between *Shipment Delay* and *Facility Halt* suggests the need for richer annotation strategies, hierarchical taxonomies, or adversarial training.

B.6.5 Document Segmentation

Fixed, non-overlapping chunks simplify inference but may fragment contextual information across document boundaries. Future work should investigate semantic and context-aware chunking strategies.

B.6.6 Multi-Event Reasoning

Large documents frequently describe multiple related events that are currently extracted independently. Event deduplication, temporal reasoning, and cross-chunk entity resolution are promising directions for improving document-level reasoning.

B.6.7 Pipeline Engineering Beyond Model Selection

The primary lesson from this work is that reliable enterprise information extraction depends on end-to-end pipeline engineering rather than language model selection alone. Dataset preparation, staged inference, structured decoding, validation, and deployment strategy collectively contributed more to system performance than changes to the underlying foundation model. Future work should continue improving these engineering methodologies to enable reliable and resource-efficient enterprise deployments.